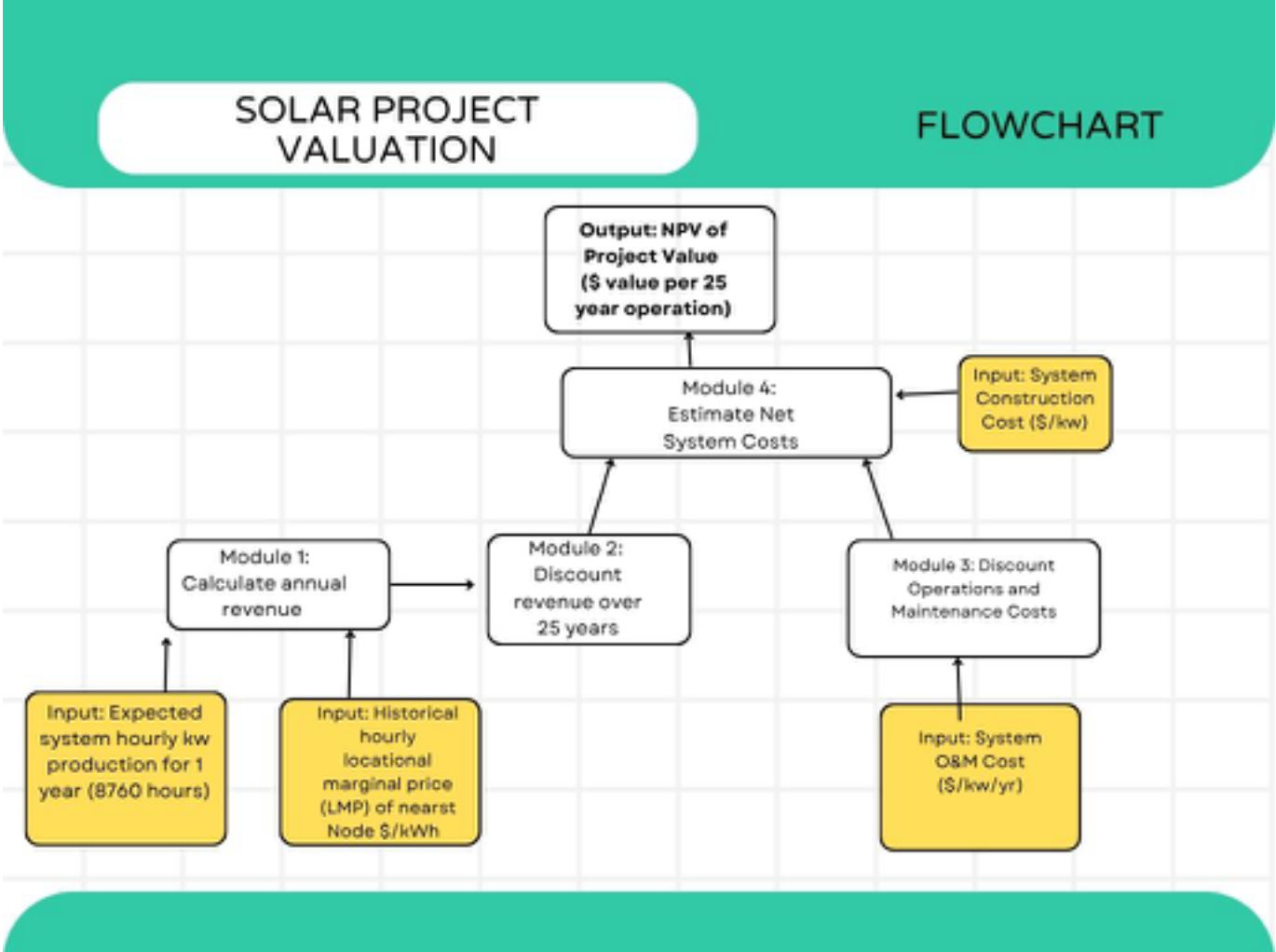


5-assignment

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Solar Project Valuation

Planning



Module Contract

Overall contract:

Input:

- year, total years for project valuation
- rate, discount rate
- size, project size (kw)
- om, operations and maintenance costs (\$/kw/yr)
- construction, year 0 costs to build (\$/kw)

Function:

- Module 1: Calculate expected annual revenue
- Module 2: Discount revenue over project lifespan in years
- Module 3: Discount operations and maintenance costs
- Module 4: Sum all costs and revenue for total project value

Final Output: Net present value of the project over total project years (single \$ value)

Module 1: Calculate expected annual revenue

Input:

- Historical hourly locational marginal price (LMP) of nearest node rates for previous year (\$/kWh)
- Hourly power generation (kW) estimate for the latitude and longitude of selected site from SAM

Function: annual_rev = Sum(hourly kW * (\$/KW) [use For loop to sum (8760 values)]

Output: Annual revenue estimate in \$/year

Module 2: Discount Revenue over project lifespan (years)

Input:

- Module 1 output (annual_rev_result)

Parameters:

- rate (Discount Rate)
- year

Function: dis_rev = annual_rev/ (1 + rate) ^ year

Output: dis_rev, discounted revenue as single \$ value across project timeline

Module 3: Discount Operations and Maintenance Costs

Input:

- om (System operations and maintenance cost (\$/kw/yr))

Parameters:

- rate (Discount Rate)
- year

Function: dis_om= (size * om)/(1+rate)^year

Output: dis_om_result, discounted operations and maintenance cost as single \$ value across project timeline

Module 4: Estimate Net System Costs

Input:

- Module 2 Output: dis_rev_result
- Module 3 Output: dis_om_result
- Construction (System Construction Cost in year 0 (\$/kw))

Function: net_npv = dis_rev_result - construction - dis_om_result

Final Output: Net present value of the project over total project years (single \$ value)

Calling functions

```
# Load Data
solardf <- read.csv("solardata.csv")
library(tidyverse)

# Clean Data
solardf_clean <- solardf |>
  rename(time = Time.stamp, lmp = Dollar.per.kwh, kwh = System.power.generated..kW.)
```

```
# Project Parameters
params <- list(
  year = 25,
  rate = 0.05,
  size = 100000,
  om = 20.39,
  construction = 1432
)
```

Module 1 Call

```
source(here("5-assignment/1_module.R"))
annual_rev_result <- annual_rev(solardf_clean$lmp, solardf_clean$kwh)
annual_rev_result
```

[1] 7587896

Module 2 Call

```
source(here("5-assignment/2_module.R"))
dis_rev_result <- dis_rev(annual_rev_result, params$year, params$rate)
dis_rev_result
```

[1] 106943379

Module 3 Call

```
source(here("5-assignment/3_module.R"))
dis_om_result <- dis_om (params$om, params$size, params$year , params$rate)
dis_om_result
```

[1] 28737553

Module 4 Call

```
source(here("5-assignment/4_module.R"))
project_npv <- net_npv(params$construction, params$size)
project_npv
```

[1] -64994174